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Task #16

Topic

React Todo App
using reusable components

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SUBMITTED TO:

APPVERSE TECHNOLOGIES

App Structure Overview

App.jsx (Main Component)

This is the root component that:

- Manages the todo list state (`todos`)
- Handles adding, updating, deleting, and toggling todos
- Loads and saves todos to `localStorage` using `useEffect`
- Wraps child components in `TodoProvider` to share the state globally

contexts/TodoContext.jsx

- Uses React's Context API to avoid prop drilling
- `TodosContext`: a shared context object with `todos` and `methods`
- `TodosProvider`: wraps the app to make context available
- `useTodo`: a custom hook to access the context in any child component

TodoForm.jsx

- Contains the input field and “Add” button
- On form submit, sends the input todo to `addTodo`
- Clears the input field after submission

TodoItem.jsx

- Renders each todo with:
 - Checkbox to mark complete
 - Editable input to update the text
 - Buttons to save, edit, and delete

- Applies conditional styling based on completion state
- Handles local state to track edit mode and updated text

Key React Concepts Used

Feature	Description
<code>useState</code>	Manages individual todo and input states
<code>useEffect</code>	Loads/saves todos to <code>localStorage</code>
<code>createContext</code> + <code>useContext</code>	Shares todos and functions across components
<code>map</code> , <code>filter</code>	Used to update or delete specific todos
<code>localStorage</code>	Persists todos across browser reloads

Tailwind CSS Highlights

- Utility-first classes for layout, spacing, colors, shadows, hover effects, etc.
- Responsive, minimal, and customizable UI
- Gradient buttons, jewel-tone backgrounds, dark theme styling